

Box FTPS Connector

for REDCap

User Training Manual

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1. Overview

The Box FTPS Connector is a REDCap External Module that automates the transfer of files from a Box account (via secure FTPS) into REDCap. It can perform five distinct operations:

- **Data Import:** Download CSV files from Box and import their rows as REDCap records.
- **File Repository Upload:** Download files from Box and place them into the REDCap File Repository.
- **Upload Field Jobs:** Match files on Box to existing REDCap records by filename and upload each file to a designated file-upload field in that record.
- **Archive Jobs:** Move files on Box from a source folder to an archive folder (without touching REDCap).
- **Delete Jobs:** Delete files from Box after they have been processed.

Each job type can be independently enabled or disabled, and multiple jobs of the same type can run in a single scheduled pass. The module runs automatically on an hourly cron schedule and can also be triggered manually through the REDCap project menu.

How the module connects

All communication with Box uses FTPS (FTP over TLS/SSL), which encrypts both commands and file data. The module requires an FTPS hostname, port, username, and password. All credentials are stored as encrypted REDCap project settings.

2. Prerequisites

Before configuring the module, confirm the following are in place :

1. **Box FTPS access:** Your Box account must have FTPS enabled. Obtain the FTPS hostname (site), port number, username, and password from your Box administrator.
2. **REDCap API token:** Generate an API token for your REDCap project (My Projects → API → Generate Token). The token must have sufficient rights for the operations you plan to run (e.g., Import Records, File Repository access).
3. **Module installed and enabled:** A REDCap administrator must have installed the Box FTPS Connector module and enabled it for your project.
4. **Folder structure on Box:** Plan your source paths and, if using Archive Jobs, your archive destination paths before configuring the module.

TLS Security Note

The module enforces TLS 1.2 and validates the Box server's SSL certificate. Self-signed certificates will cause connection failures. Contact your Box administrator if you encounter SSL errors.

3. Enabling the Module

To access module settings, navigate to:

My Projects → [Your Project] → External Modules → Box FTPS Connector → Configure

The configuration page is organized into sections. Complete them top to bottom: Connection Settings first, then Scheduling, then individual Job Types.

Status Banner

Once the module has run successfully, a color-coded status banner appears on the page for your project. A green banner means the last run succeeded.

4. Connection Settings

These settings in the config authenticate the module to Box and to the REDCap API. They apply to all job types.

Setting	Type	Required?	Description
Site	Text	Required	FTPS hostname for your Box account, e.g. ftp.box.com
Port	Text	Required	FTPS port number, typically 990 or 21. Confirm with your Box admin.
Username	Password	Required	FTPS username for Box.
Password	Password	Required	FTPS password for Box.
API Token	Password	Required	REDCap API token for this project.

Testing the Connection

After saving connection settings, use the manual run button (see Section 7) to confirm the module can reach Box. Check the Import Log section that appears on screen for any FTPS ERROR messages.

5. Scheduling

The module includes a built-in scheduler that fires the cron job every hour. Within that hourly window, you control which hours and days the actual import runs.

Enabling the Cron

Enable Cron checkbox must be checked for any automatic runs to occur. If unchecked, the module settings are saved but no automatic imports happen.

Choosing Run Times

Two run times can be configured, both in 24-hour format (0–23):

Setting	Description
Time of Day	Primary run hour. Enter a whole number, e.g. "14" for 2:00 PM.
Second Time of Day	Optional second run hour. Useful for a morning and afternoon transfer. Leave blank if only one daily run is needed.

Choosing Run Days

Check the box next to each day the import should run. All seven days are independently selectable. At minimum, one day must be checked for the scheduler to do anything.

How timing works

The REDCap cron system calls the module once per hour. The module compares the current hour to your configured times and the current day of the week to your selected days. If both match, the import runs. If either does not match, the module skips silently — nothing is logged for a skipped run.

6. Job Types

Each job type is independently enabled by its own checkbox. Multiple jobs of the same type can be added using the repeatable sub-settings. All job types execute in a single run in the following order: Data Import → File Repository Upload → Upload Field → Archive → Delete.

Wildcard Support

All file path fields support the * wildcard (matches any characters) and the wildcard (matches a single character). For example, /incoming/*.csv will match all CSV files in the /incoming/ folder. Exact paths like /incoming/data.csv are also supported.

6.1 Data Import Jobs

Downloads one or more CSV files from Box and imports their rows into REDCap as records. The first row of the CSV must be a header row.

How to Enable

- **Enable Data Import Jobs** checkbox must be checked.

Sub-Settings (repeatable)

Field	Required?	Description
Data File Path or Pattern	Required	Path or wildcard pattern on Box, e.g. /imports/records.csv or /imports/*.csv
Field Map	Optional	Column-to-field mapping in the format csv_column:redcap_field, csv_column2:redcap_field2. If left blank, CSV headers are used as-is and must exactly match REDCap variable names.

Field Map Examples

CSV headers match REDCap exactly — leave blank:

(leave empty)

Rename two columns:

PatientID:record_id, CollectionDate:collection_date

How it Works

5. The module downloads the CSV to a temporary server location.
6. It reads the header row to determine column names.
7. If a field map is provided, only mapped columns are imported; unmapped columns are ignored.
8. All rows are uploaded to REDCap via the API in a single batch call.
9. The temporary file is deleted after import.

REDCap API Rights

The API token must have Import Records permission. If your project uses events (longitudinal), the CSV must include a `redcap_event_name` column, or records will be imported to the first event.

6.2 File Repository Upload Jobs

Downloads files from Box and uploads them into the REDCap File Repository, optionally into a specific folder.

How to Enable

- **Enable File Repository Upload Jobs** checkbox must be checked.

Sub-Settings (repeatable)

Field	Required?	Description
Repository Files Pattern	Required	Path or wildcard pattern on Box.
Repo Folder ID	Required	Numeric ID of the REDCap File Repository folder. Find this by hovering over a folder in the File Repository; the ID appears in the URL.
Overwrite files with same name	Required	If checked, an existing file with the same name in the target folder is deleted before the new file is uploaded. If unchecked, a duplicate will be created.

Finding a Folder ID

10. Open your REDCap project.
11. Go to File Repository.
12. Navigate to or create the target folder.
13. Click the folder to open it and look at the browser URL bar.
14. Copy the numeric value of the `folder_id` parameter from the URL.

6.3 Upload Field Jobs

Matches files on Box to specific REDCap records based on filename conventions, and uploads each file to a designated file-upload field within that record.

How to Enable

- **Enable Upload Field Jobs** checkbox must be checked.

Filename Convention

The filename on Box determines which record receives the file. The module splits the filename on the configured delimiter and uses the first segment as the Record ID.

Example: Delimiter is underscore (_), filename is **1001_consent.pdf**. The module looks for Record ID **1001** and, if found, uploads the full file.

If no matching record is found, the file is moved to the configured Error Directory on Box.

Sub-Settings (repeatable)

Field	Required?	Description
Source Directory on Box	Required	Folder path on Box to scan for files, e.g. /Consents/Incoming
REDCap File Upload Field	Required	Select the REDCap field (must be a File Upload field type) from the dropdown.
REDCap Event	Optional	For longitudinal projects, enter the unique event name (not the label) where the file upload field lives. Leave blank for classic projects.
Filename Delimiter	Required	Character that separates the Record ID from the rest of the filename. Choose from underscore, hyphen, period, or space.
Error Directory on Box	Required	Box folder path where files are moved when no matching REDCap record is found. Acts as a quarantine folder.

6.4 Archive Jobs

Moves files on Box from a source location to an archive folder. No data is imported into REDCap. This is useful for post-processing cleanup or maintaining a file history.

How to Enable

- **Enable Archive Jobs** checkbox must be checked.

How it Works

15. The module downloads the source file to a temporary location on the REDCap server.
16. It uploads the file to the archive destination on Box.
17. Only if the upload succeeds, the original file is deleted from Box.
18. The temporary local file is always cleaned up.

Safety Note

The original file is only deleted after a confirmed successful upload to the archive destination. If the archive upload fails, the original remains in place and an error is logged.

Sub-Settings (repeatable)

Field	Required?	Description
Source File Path or Pattern	Required	Path or wildcard pattern of files to archive on Box.
Archive Folder Path on Box	Required	Destination folder on Box where archived files will be placed, e.g. /imports/archive

6.5 Delete Jobs

Permanently deletes files from Box. Typically used after files have been imported or archived to clean up the source directory.

Warning — Permanent Action

Deletion cannot be undone. Always pair Delete Jobs with an Archive Job or confirm the files are no longer needed before enabling deletion.

How to Enable

- **Enable Delete Jobs** checkbox must be checked.

Sub-Settings (repeatable)

Field	Required?	Description
File Path or Pattern to Delete	Required	Path or wildcard pattern of files to delete from Box.

7. Running an Import Manually

The module can be triggered immediately without waiting for the scheduled cron, which is useful for testing or on-demand transfers.

How to Run Manually

19. Open your REDCap project.
20. In the left sidebar, click **BoxConnector** under External Modules.
21. The module runs immediately and displays a live log on screen.

The on-screen log uses color coding:

- **Green text:** File downloaded, records imported, or other success events.
- **Orange text:** Warnings that did not stop the run.
- **Red text:** Errors that caused a job step to fail.

Note on Scheduling

Running manually does NOT check the schedule settings. It always executes regardless of the time or day configured. This is intentional so you can test at any time.

8. Monitoring & Notifications

8.1 Error Log

The Error Log setting (visible in the module configuration page) stores the two most recent run logs. When errors occur, the full log is preserved. When runs succeed, only info and summary messages are kept. This rolling log is automatically maintained by the module.

8.2 Email Notifications

Notification Email (optional) — Enter one email address in this field. The module will send an email notification any time a cron run encounters errors. The email includes:

- The project ID and run timestamp.
- The specific error messages that occurred.
- A note that the full log is available in module settings.

Only error runs trigger email

Successful runs do not send email. If you want to verify a run happened, check the status banner or the Error Log in module settings.

9. Troubleshooting

Symptom	What to Check
FTPS ERROR in log	Verify Site, Port, Username, and Password. Confirm Box FTPS is enabled on your account. Check that the REDCap server can reach the Box FTPS hostname on the configured port.
REDCap API Error in log	Verify the API Token is correct and has not expired. Confirm the token has the necessary rights (Import Records, File Repository) in REDCap.
No files matched pattern	Confirm the file path exists on Box and the pattern is correct. Remember paths are case-sensitive. Wildcards (* and ?) are supported; dots and extensions must be included.
No job types enabled	At least one of the Enable checkboxes (Data Import, Repo Upload, Upload Field, Archive, Delete) must be checked or the module will exit immediately.
Cron never runs	Verify Enable Cron is checked. Confirm at least one day checkbox is selected and the configured hour matches when you expect a run. Note: time is based on the server's local time zone.
File moved to Error Directory (Upload Field)	The Record ID extracted from the filename did not match any existing record in REDCap. Verify the filename delimiter is correct and the record exists. Check log for the specific record ID that was not found.
CSV imports fail or missing fields	If no field map is provided, CSV column names must exactly match REDCap variable names (case-sensitive). Check for leading/trailing spaces in column headers.
Invalid folder ID for repo job	The Repo Folder ID must be a positive whole number. Verify you copied the correct numeric ID from the REDCap File Repository URL.

10. Quick-Reference Summary

Execution Order

When a run is triggered (cron or manual), jobs execute in this sequence:

- 22. **Data Import Jobs** — CSV → REDCap records
- 23. **File Repository Upload Jobs** — Files → REDCap File Repository
- 24. **Upload Field Jobs** — Files → Individual record upload fields
- 25. **Archive Jobs** — Move files within Box
- 26. **Delete Jobs** — Delete files from Box

Wildcard Cheatsheet

Pattern	Example	Matches
*	/incoming/*.csv	All .csv files in /incoming/
?	/data/record?.csv	record1.csv, record2.csv, etc.
exact path	/data/export.csv	Only that specific file

24-Hour Time Reference

Enter this value	Runs at
0	12:00 AM (midnight)
6	6:00 AM
8	8:00 AM
12	12:00 PM (noon)
14	2:00 PM
17	5:00 PM
20	8:00 PM
23	11:00 PM

Checklist: First-Time Setup

- 27. **Enable module:** Confirm the module is enabled for your project by a REDCap admin.
- 28. **Connection:** Enter Site, Port, Username, Password, and API Token.
- 29. **Schedule:** Check Enable Cron, set at least one time, and select at least one day.

- 30. **Jobs:** Enable at least one job type and configure its sub-settings.
- 31. **Test:** Run manually via the BoxConnector link and review the on-screen log.
- 32. **Notification:** Optionally add an email address for error alerts.

Support

For questions about this module, contact your REDCap system administrator or the module author: Gary Rains (gary.rains@duke.edu), Duke University School of Medicine.